CRAWDAD 2026 · EVALUATION BRIEF

Measure what holds a text together.

Centering Resonance Analysis for research teams — a published method, a deterministic implementation, and nothing to install.

- Runs offline in a browser
- Byte-identical reruns
- Method published 2002, cited in every output

Version 2.0.1 · 2026-10-28

01 · THE METHOD

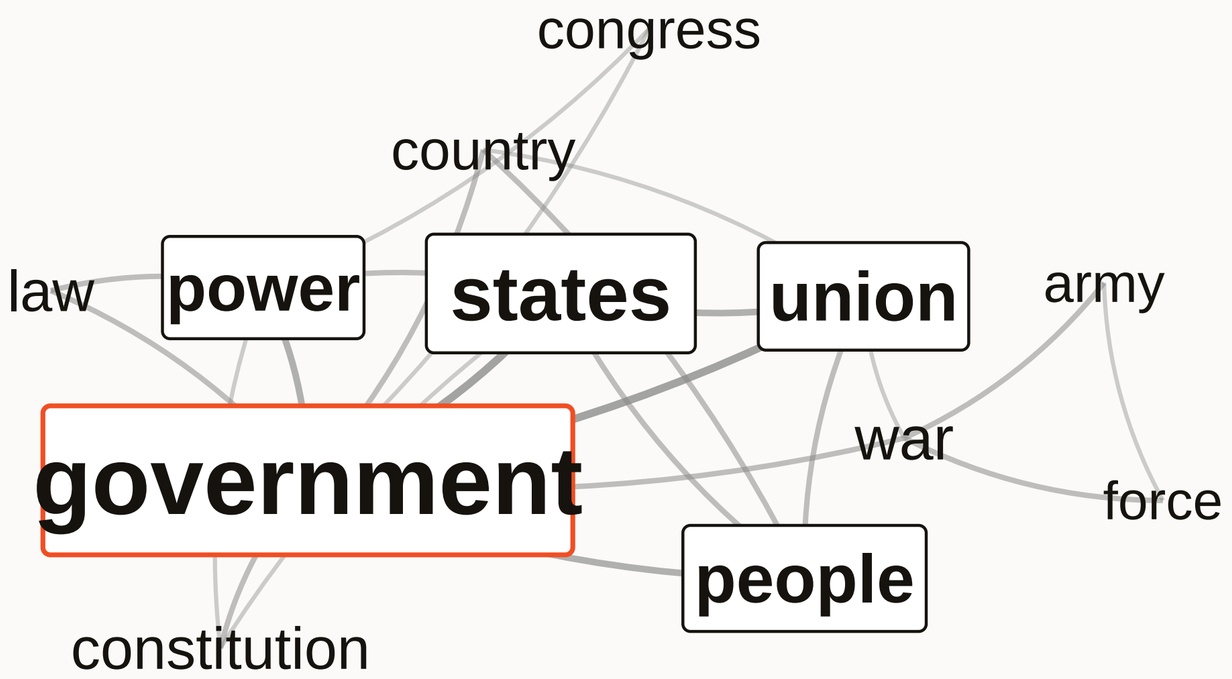
Influence is not frequency

CRA takes the nouns and adjectives inside noun phrases, links the ones adjacent in discourse, and scores each word by how often it lies on the shortest path between other words. The result is the vocabulary doing the organising work — not the loudest vocabulary.

01 Answers questions counting cannot: what organises this text, how close are two texts, how does a corpus group.

02 No model, no training data, no prompt — a defined calculation you can explain to a reviewer.

03 More than twenty text-level metrics come along for free — sentiment (AFINN and VADER), readability, formality, diversity, cohesion, and an information-theoretic profile.



Lincoln’s message to Congress, July 1861 — twelve words from its 427-word network, placed by their links. Orange box: highest influence. Black boxes: influence above 0.05.

02 · SCOPE

One conversion. Ten analyses.

The Generator is the only step that reads your documents. It writes one plain-markdown network file per text; every module below reads those files — so a completed study is a folder you can archive, email, or hand to a reviewer.

Visualizer

The network drawn in the original Crawdad style, exported at 1920×1080 for slides.

`1+ file`

Comparator

Four resonance values between two texts, with their shared and unique influential words.

`2 files`

Classifier

Seeded K-means over mutual resonance, plus a hierarchical dendrogram you cut yourself.

`5+ files`

Theme Identifier

PCA with varimax rotation over the corpus influence matrix, with defining words per theme.

`5+ files`

Latent Themes

Scores texts against your own coding dictionary, and reports every code word that matched nothing.

`1+ file`

Summarizer

One row per text, 29 columns — a flat dataset ready for R, Stata or SPSS.

`2+ files`

Keyness

Two groups of texts compared word by word: G^2 evidence, Log-Ratio effect sizes, dispersion warnings, Zeta markers.

`2+ files per group`

Topic Modeler

Deterministic NMF topics with a coherence-guided topic count — no random seeds to hide.

`8+ files`

Stylometry

Burrows and Cosine Delta style distances with clustering for free, and an optional compression cross-check.

`3+ files`

Dynamics

The published windowed-CRA workflow: dynamic themes, change points, turnover, novelty over time.

`6+ windows`

03 · DEFENSIBILITY

Deterministic, and audited

The same files with the same settings produce byte-identical results on any machine. There is no model, no randomness — clustering uses a fixed seed — and no network dependence. Every output file records the parameters that produced it.

A nine-component validation study checked the core calculations against NetworkX, scikit-learn, SciPy and hand-worked examples, and the golden corpus still reproduces byte-identically. Version 2.0 extends the oracle battery: every new method is validated against an independent reference implementation — VADER, gensim, statsmodels, ruptures, leidenalg, scikit-learn NMF — or a hand-worked closed form, with seeds and counts recorded.

0.975

Parser F1 against a 60-sentence hand-annotated gold corpus

291

Automated checks you can run yourself in `tests/index.html`

SHA-256

A published checksum for every file in the distribution

2002

Peer-reviewed method, cited in every output file

04 · ADOPTION COST

Unzip. Double-click. Done.

There is no server to stand up, no licence server, no account, and no vendor between you and your data. The security review is a folder of files and a checksum list.

Nothing is installed

Unzip anywhere and open `index.html`. No administrator rights. Chrome, Edge, Firefox or Safari, on Mac, Windows or Linux.

Nothing leaves the machine

No network access, no storage, no telemetry — documented in `SECURITY-AND-PROVENANCE.md` along with every bundled open-source licence.

Works with what you have

Ingests .txt, .md, .pdf and .docx, cleans citation artifacts automatically, and logs exactly what it removed from each file.

Outputs you already know

Markdown or PDF reports, slide-ready PNGs, machine-readable tables, and a one-click session bundle that moves a whole study to another machine.

05 · DOES IT ACTUALLY WORK

It recovers structure it was never told about

Given 26 Lincoln texts spanning 1832–1865 and no dates, titles or genre labels, the Classifier isolated the 1857–58 debate speeches as one cluster and the late-war canon — Gettysburg, the last two annual messages, the Second Inaugural — as another.

The Theme Identifier independently produced a debate theme defined by *slavery, judge, question, territories* and a constitutional-crisis theme defined by *laws, government, union, constitution*. Coherence against information the algorithm never saw is the strongest validity check available.

A second test in 2.0: Dynamics re-ran the published Iridium news-stream design (164 WSJ articles, 1990–2002) — identical themes and change points to the pilot analysis, agreeing with the independently hand-coded 2007 PICMET study to within a year.

Corman, S., Kuhn, T., McPhee, R., & Dooley, K. (2002). Studying complex discursive systems: Centering Resonance Analysis of organizational communication. *Human Communication Research, 28*(2), 157–206.
Dooley, K. (2026). *Crawdad Text Analysis Software* (Version 2.0.1) [Computer software].

WHAT IT WILL NOT DO FOR YOU

It will not interpret. The software measures structure; building the argument, and standing behind it, remains your job.

It will not rescue short texts. Under roughly 100 nodes, metrics are unstable and clusters group by sparsity rather than content.

It will not bridge synonyms. Resonance matches stems; two texts on one topic in different vocabulary score low, correctly.

Two metrics are heuristics. Abstractness and agreeableness are transparent proxies, flagged as such, for comparison only.